الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

WEEK 8 · LESSONS 7-2 · 7-3 · 7-4 · 7-5

Weekly Quiz Preparation

Grade 11 Advanced · Mathematics

كتيّب التجهّز للاختبار الأسبوعي القصير

1 2 3 4

4	10	40	12
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

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One page per objective · every solution written out step by step

Quiz map

4 objectives = 4 recipes · 4 أهداف = 4 وصفات

1 **Complex fractions** الكسور المركبة
Top as one fraction · bottom as one fraction · multiply by the reciprocal · cancel

2 **Graph a/(x – h) + k** تحويلات الدالة
Dashed x = h and y = k first · two points either side · domain x ≠ h, range y ≠ k

3 **Oblique asymptotes** التقارب المائل
Zeros: top = 0 · vertical: bottom = 0 · oblique: the quotient of the division

4 **Direct and joint variation** التغير الطردي والمشترك
y = kx or a = kbc · find k from the first values · then substitute

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 Complex fractions
الكسور المركبة

A complex fraction is just a division: write the top as ONE fraction, the bottom as ONE fraction, then divide.

Combine the numerator into one fraction and the denominator into one fraction (common denominators), then multiply by the reciprocal and cancel.

اكتب البسط كسرًا واحدًا والمقام كسرًا واحدًا بتوحيد المقامات، ثم اضرب في المقلوب واختصر.

Recipe الوصفة

1. **Top and bottom: common denominator, add or subtract**
   البسط والمقام: وحّد المقامات
2. **Top ÷ bottom = top × reciprocal of the bottom**
   اقسم: اضرب في المقلوب
3. **Cancel the common factors**
   اختصر العوامل المشتركة

Watch out

- Subtract the WHOLE second numerator: −(x − 1) changes both signs.

Worked example Authored (shape of Ex 16-19)

Simplify $\dfrac{\dfrac{1}{x+1}+\dfrac{2}{x-1}}{\dfrac{3}{x-1}-\dfrac{1}{x+1}}$.

1. LCD (x+1)(x−1): add the numerators

$$\text{top} = \frac{(x-1)+2(x+1)}{(x+1)(x-1)} = \frac{3x+1}{(x+1)(x-1)}$$

2. same LCD: subtract the numerators (both terms of −(x−1))

$$\text{bottom} = \frac{3(x+1)-(x-1)}{(x+1)(x-1)} = \frac{2x+4}{(x+1)(x-1)}$$

3. multiply by the reciprocal, cancel (x+1)(x−1)

$$\frac{3x+1}{(x+1)(x-1)} \cdot \frac{(x+1)(x-1)}{2x+4} = \frac{3x+1}{2x+4}$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 16) Simplify $\dfrac{\dfrac{3}{x+2}+\dfrac{1}{x-2}}{\dfrac{2}{x-2}-\dfrac{1}{x+2}}$.
2. Authored (shape of Ex 18) Simplify $\dfrac{\dfrac{2}{x}+\dfrac{3}{x+1}}{\dfrac{1}{x}-\dfrac{2}{x+1}}$.
3. Authored (shape of Ex 19) Simplify $\dfrac{\dfrac{4}{x-3}-\dfrac{5}{x+3}}{\dfrac{1}{x-3}+\dfrac{2}{x+3}}$.

2 Transformations of reciprocal functions
تحويلات الدوال النسبية الأساسية

Read h and k from the equation, draw the two dashed asymptotes first, then plot points either side.

f(x) = a/(x − h) + k has the vertical asymptote x = h and the horizontal asymptote y = k. Domain: x ≠ h. Range: y ≠ k.

للدالة f(x) = a/(x − h) + k خط تقارب رأسي x = h وأفقي y = k؛ المجال x ≠ h والمدى y ≠ k.

Recipe الوصفة

1. **Match the form a/(x − h) + k: read a, h, k**
   اقرأ a وh وk
2. **Dashed lines x = h and y = k**
   ارسم خطي التقارب بخط متقطع
3. **Two points either side of x = h**
   نقطتان على جانبي x = h
4. **a > 0: upper right and lower left; a < 0: upper left and lower right**
   إشارة a تحدد الفرعين

Watch out

- x + 3 means h = −3 (the sign flips), but +k keeps its sign.
- The graph never touches either dashed line.

Tool. asymptotes: x = h and y = k

Worked example Authored (shape of Ex 13-16)

Graph $f(x) = \dfrac{2}{x-3} + 1$. State the asymptotes, the domain and the range.

1. match a/(x − h) + k

   $a = 2,\ h = 3,\ k = 1$

2. asymptotes, domain, range

   $x = 3$ and $y = 1$ (dashed); domain $x \neq 3$, range $y \neq 1$

3. substitute x either side of 3

   $(4, 3),\ (5, 2),\ (2, -1),\ (1, 0)$

4. two branches that never touch the dashed lines

   $a > 0$: upper right and lower left branches

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 13) Graph $f(x) = \dfrac{1}{x-2} + 3$. State the asymptotes, the domain and the range.
2. Authored (shape of Ex 14) Graph $f(x) = \dfrac{-2}{x+1} - 4$. State the asymptotes, the domain and the range.
3. Authored (shape of Ex 16) Graph $f(x) = \dfrac{3}{x+6} + 2$. State the asymptotes, the domain and the range.

3 Oblique asymptotes

خطوط التقارب المائلة

When the top is one degree higher than the bottom, long division gives the slanted line the graph follows.

Zeros: the numerator is 0. Vertical asymptote: the denominator is 0 (the numerator is not). Oblique asymptote: divide — the quotient is the line y = mx + b.

الأصفار: البسط = 0. التقارب الرأسي: المقام = 0. المائل: اقسم، والناتج (دون الباقي) هو المستقيم y = mx + b.

Recipe الوصفة

1. **Zeros: set the numerator = 0**
   الأصفار: البسط = 0
2. **Vertical asymptote: set the denominator = 0**
   التقارب الرأسي: المقام = 0
3. **Divide the numerator by the denominator**
   اقسم البسط على المقام
4. **The quotient is the oblique asymptote; the remainder is dropped**
   الناتج هو المائل، ويُهمل الباقي

Watch out

- Expand $(x + 2)^2$ before dividing: $x^2 + 4x + 4$.
- The remainder fraction tends to 0: it is NOT part of the asymptote.

Worked example Authored (shape of Example 4)

Find the zero, the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{(x+2)^2}{x-1}$.

1. numerator 0 → zero; denominator 0 → vertical asymptote

   zero: $(x+2)^2 = 0 \Rightarrow x = -2$; vertical asymptote: $x - 1 = 0 \Rightarrow x = 1$

2. long division

   $(x^2 + 4x + 4) \div (x - 1) = x + 5$, remainder 9

3. the remainder fraction tends to 0

   $$f(x) = x + 5 + \frac{9}{x-1}$$

4. the quotient is the line

   oblique asymptote: $y = x + 5$

Your turn · solve in your notebook · solutions at the back

1. **Authored (shape of Ex 11)** Find the zero, the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{(x-1)^2}{x+3}$.
2. **Authored (shape of Ex 14)** Find the zeros, the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{2x^2 + 3x}{x-1}$.
3. **Authored (shape of Ex 16)** Find the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{x^2+1}{x+2}$. Has it any zero?

4 Direct and joint variation

التغير الطردي والمشترك

Variation is a formula with a constant k: find k from the first set of values, then use it.

Direct: y = kx. Joint: a = kbc (a varies with the PRODUCT of b and c). Use the given values to find k, then substitute the new values.

طردي: y = kx. مشترك: a = kbc. أوجد k من القيم المعطاة ثم عوّض بالقيم الجديدة.

Recipe الوصفة

1. **Write the formula: y = kx or a = kbc** — اكتب الصيغة
2. **Put the first set of values in: find k** — عوّض لإيجاد k
3. **Put k and the new values in** — عوّض بـ k والقيم الجديدة
4. **Calculate; keep the signs** — احسب وانتبه للإشارات

Watch out

- k is found ONCE, from the first set of values.
- Keep the minus signs: (−5)(4) = −20.

Tool. direct: y = kx · joint: a = kbc

Worked example Authored (shape of Ex 1-12)

y varies directly as x, and $y = 12$ when $x = 4$. Find y when $x = 9$. Then: z varies jointly as x and y, and $z = 30$ when $x = 2$ and $y = 3$; find z when $x = 4$ and $y = 5$.

1. direct variation: find k, then substitute

$y = kx$: $12 = k(4) \Rightarrow k = 3$, $y = 3(9) = 27$

2. joint variation: z = kxy, find k

$z = kxy$: $30 = k(2)(3) \Rightarrow k = 5$

3. substitute x = 4, y = 5

$z = 5(4)(5) = 100$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 3) p varies directly as q, and $p = -20$ when $q = 4$. Find p when $q = -7$.
2. Authored (shape of Ex 8) a varies jointly as b and c, and $a = 36$ when $b = 3$ and $c = 4$. Find a when $b = 5$ and $c = 2$.
3. Authored (shape of Ex 11) y varies jointly with x and z, and $y = -24$ when $x = 2$ and $z = -6$. Find y when $x = -3$ and $z = 4$.

Objective 1 · Complex fractions الكسور المركبة

1.1 Authored (shape of Ex 16)

Simplify $\dfrac{\dfrac{3}{x+2}+\dfrac{1}{x-2}}{\dfrac{2}{x-2}-\dfrac{1}{x+2}}$.

1. LCD $(x+2)(x-2)$

$$\text{top} = \frac{3(x-2)+(x+2)}{(x+2)(x-2)} = \frac{4x-4}{(x+2)(x-2)}$$

2. subtract the whole second numerator

$$\text{bottom} = \frac{2(x+2)-(x-2)}{(x+2)(x-2)} = \frac{x+6}{(x+2)(x-2)}$$

3. cancel $(x+2)(x-2)$

$$= \frac{4x-4}{x+6}$$

1.2 Authored (shape of Ex 18)

Simplify $\dfrac{\dfrac{2}{x}+\dfrac{3}{x+1}}{\dfrac{1}{x}-\dfrac{2}{x+1}}$.

1. LCD $x(x+1)$

$$\text{top} = \frac{2(x+1)+3x}{x(x+1)} = \frac{5x+2}{x(x+1)}$$

2. the same LCD

$$\text{bottom} = \frac{(x+1)-2x}{x(x+1)} = \frac{1-x}{x(x+1)}$$

3. cancel $x(x+1)$

$$= \frac{5x+2}{1-x}$$

1.3 Authored (shape of Ex 19)

Simplify $\dfrac{\dfrac{4}{x-3}-\dfrac{5}{x+3}}{\dfrac{1}{x-3}+\dfrac{2}{x+3}}$.

1. LCD $(x-3)(x+3)$

$$\text{top} = \frac{4(x+3)-5(x-3)}{(x-3)(x+3)} = \frac{27-x}{(x-3)(x+3)}$$

2. the same LCD

$$\text{bottom} = \frac{(x+3)+2(x-3)}{(x-3)(x+3)} = \frac{3x-3}{(x-3)(x+3)}$$

3. cancel $(x-3)(x+3)$

$$= \frac{27-x}{3x-3}$$

Objective 2 · Transformations of reciprocal functions — تحويلات الدوال النسبية الأساسية

2.1 Authored (shape of Ex 13)

Graph $f(x) = \dfrac{1}{x-2}+3$**. State the asymptotes, the domain and the range.**

1. read a, h, k

$a = 1,\ h = 2,\ k = 3:\ x = 2,\ y = 3$ (dashed); $x \in \mathbb{R} \setminus \{2\},\ y \in \mathbb{R} \setminus \{3\}$

2. points either side of $x = 2$

$(3, 4),\ (1, 2)$ and $(0, 2.5)$

3. two branches

$a > 0$: upper right, lower left

2.2 Authored (shape of Ex 14)

Graph $f(x) = \dfrac{-2}{x+1}-4$**. State the asymptotes, the domain and the range.**

1. $x + 1$ means $h = -1$

$a = -2,\ h = -1,\ k = -4:\ x = -1,\ y = -4$ (dashed); $x \in \mathbb{R} \setminus \{-1\},\ y \in \mathbb{R} \setminus \{-4\}$

2. points either side of $x = -1$

$(0, -6),\ (-2, -2)$

3. a is negative: the branches swap

$a < 0$: upper left, lower right

2.3 Authored (shape of Ex 16)

Graph $f(x) = \dfrac{3}{x+6} + 2$. State the asymptotes, the domain and the range.

1. x + 6 means h = −6

   $a = 3,\ h = -6,\ k = 2:\ x = -6,\ y = 2$ (dashed); $x \in \mathbb{R} - \{-6\},\ y \in \mathbb{R} - \{2\}$

2. points either side of x = −6

   $(-5, 5),\ (-7, -1)$

3. two branches

   $a > 0$: upper right, lower left

Objective 3 · Oblique asymptotes — خطوط التقارب المائلة

3.1 Authored (shape of Ex 11)

Find the zero, the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{(x-1)^2}{x+3}$.

1. numerator 0 and denominator 0

   zero: $x = 1$; vertical asymptote: $x = -3$

2. long division

   $(x^2 - 2x + 1) \div (x + 3) = x - 5$, remainder 16

3. the quotient is the line

   oblique asymptote: $y = x - 5$

3.2 Authored (shape of Ex 14)

Find the zeros, the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{2x^2 + 3x}{x - 1}$.

1. factor the numerator

   $2x^2 + 3x = x(2x + 3) = 0:\ x = 0,\ x = -\frac{3}{2}$; vertical asymptote: $x = 1$

2. long division

   $(2x^2 + 3x) \div (x - 1) = 2x + 5$, remainder 5

3. the quotient is the line

   oblique asymptote: $y = 2x + 5$

3.3 Authored (shape of Ex 16)

Find the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{x^2 + 1}{x + 2}$. Has it any zero?

1. denominator 0

   vertical asymptote: $x = -2$

2. long division

   $(x^2 + 1) \div (x + 2) = x - 2$, remainder 5

3. $x^2 + 1$ is never 0: no zero

   oblique asymptote: $y = x - 2$; $x^2 + 1 = 0$ has no real zero

4.1 Authored (shape of Ex 3)

p varies directly as q, and $p = -20$ when $q = 4$. Find p when $q = -7$.

1. direct variation: find k

$p = kq$: $-20 = k(4) \Rightarrow k = -5$

2. substitute q = −7

$p = -5(-7) = 35$

4.2 Authored (shape of Ex 8)

a varies jointly as b and c, and $a = 36$ when $b = 3$ and $c = 4$. Find a when $b = 5$ and $c = 2$.

1. joint variation: find k

$a = kbc$: $36 = k(3)(4) \Rightarrow k = 3$

2. substitute b = 5, c = 2

$a = 3(5)(2) = 30$

4.3 Authored (shape of Ex 11)

y varies jointly with x and z, and $y = -24$ when $x = 2$ and $z = -6$. Find y when $x = -3$ and $z = 4$.

1. joint variation: find k

$y = kxz$: $-24 = k(2)(-6) \Rightarrow k = 2$

2. substitute x = −3, z = 4

$y = 2(-3)(4) = -24$

Ready?

Tick each line · ضع علامة

- [ ] **I can simplify a complex fraction**
  أستطيع تبسيط الكسر المركب
- [ ] **I can graph a reciprocal function and state its domain and range**
  أستطيع تمثيل الدالة وكتابة مجالها ومداها
- [ ] **I can find zeros, vertical and oblique asymptotes**
  أستطيع إيجاد الأصفار وخطوط التقارب
- [ ] **I can solve direct and joint variation problems**
  أستطيع حل مسائل التغير الطردي والمشترك

On the day

Write the first line at once: the first step already earns a mark. In a graph, draw the dashed asymptotes before anything else.

Every question

Show the common denominator and the cancelled factor; show the long division for an oblique asymptote.

One mark for every written step

لكل خطوة تكتبها علامة